THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 6: **Heat Transfer in Nature** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Why are the radiators of vehicles painted black? 1

(a) looks nice **(b)** cheap
(c) good reflector **(d)** good emitter of heat

2. Which of these will keep ice from melting longest? 1

(a) ice in the open **(b)** ice in aluminium foil
(c) ice wrapped in thick wool **(d)** ice in a steel plate

3. Which part of a room is warmest? 1

(a) middle **(b)** near the floor **(c)** outside **(d)** near the ceiling

4. Why is aluminium foil used to wrap hot food? 1

(a) adds heat **(b)** absorbs
(c) reduces radiation and holds heat **(d)** cools

5. A metal spoon and a wooden spoon in hot water: the metal spoon heats up faster because: 1

(a) wood is better **(b)** wood is heavier
(c) metal is lighter **(d)** metal is a better conductor

6. Air conditioners are fixed high on the wall because: 1

(a) it saves wiring **(b)** warm air sinks and cools the room
(c) it looks good **(d)** cold air sinks and cools the room

7. Why do we keep a lid on a pan while cooking? 1

(a) to reduce heat loss **(b)** to cool food
(c) to add heat **(d)** to stop smell

8. A vacuum flask reduces heat loss mainly because: 1

(a) it is coloured
(b) vacuum prevents conduction and convection
(c) it is made of steel
(d) it is heavy

9. In a refrigerator, why is the freezer compartment at the top? 1

(a) cold air rises **(b)** it saves electricity
(c) cold air sinks to cool below **(d)** the door is heavy

10. Why do we prefer thermocol boxes for ice cream? 1

(a) it is hard **(b)** thermocol is an insulator

(c) it is a conductor **(d)** it is shiny

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do trucks carrying ice-cream have white shiny covers? 2

12. Why does a land breeze blow towards the sea at night? 2

13. Why do penguins huddle together? 2

14. Why do thick walls keep houses cool in summer? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Why is it advised to wear several thin layers in cold? 3

16. What are insulating materials used in refrigerators and why? 3

17. Why does a shiny steel kettle stay hot longer than a dull black one? 3

OR

Why does the water in a pond remain cool at the bottom in summer? 3

18. Why do we feel the air cooler near a lake in the evening? 3

OR

Why do objects in the shade stay cooler? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Compare heat transfer in solids, liquids and gases, using examples. 5

OR

Explain how forest fires spread and relate it to heat transfer. 5

20. Explain how a hot water geyser, an iron and a refrigerator use properties of heat transfer. 5

OR

Describe how an igloo, a mud house and a hill-station house use insulation. 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND